

VDS/DVP/BH26 Variant Branches and Coupled-Field Derivations: Session 202 T61-T80 QCalcGeom Phase H202

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PAPER_1151 -- VDS/DVP/BH26 Variant Branches and Coupled-Field Derivations: Session 202 Phase H202

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Framework: UQFF v5.77 -- Star-Magic Physics

Source: QCalcGeom.py v2.1.0 Phase H202 (commit 6e69e2f5, May 5 2026)

Integration Date: May 6, 2026 (Session 234)

Tests: T61--T70 (C++) / T71--T80 (Python); 80/80 PASS

Classification: QCalcGeom Branch Derivations -- Calibration Coupling

$$\text{joint_coeff} = \sqrt{w_{\text{VDS}} \times w_{\text{DVP}}}, \quad w_{\text{VDS}} = \frac{\text{Li}_{26}(\text{SSq})}{\text{Li}_{25}(\text{SSq})}, \quad w_{\text{DVP}} = \frac{\zeta_{\text{DVP}}(\text{SSq})}{a(29)}$$

Abstract

This paper registers the Phase H202 QCalcGeom extension (Session 202) as PAPER_1151, formally deriving the **variant branch decompositions** of the three canonical UQFF number systems -- VDS (Vacuum Density Series), DVP (Dipole Vortex Primes), and BH26 (26-dimensional Black Hole spectral ladder) -- and their joint coupling field.

The derivation produces five new result structures: VDSBranchResult, DVPBranchResult, BH26BranchResult, VDSDVPCoupledResult, and BH26BSHResonanceResult.

All 80 branch tests (T61--T80) pass at $\text{SSq} = 0.57$.

1. Background: Three Canonical Number Systems

The UQFF framework (Star-Magic.txt Chapter 4) operates three vacuum-number systems in simultaneous registers:

System	Definition	Role
VDS	$\text{Li}_{26}(z) = \sum_{n=1}^{\infty} z^n / n^{26}$	Vacuum energy series
DVP	$\zeta_{\text{DVP}}(z) = \sum_{p>26} z^{\pi(p)} / p^{26}$	Prime-vortex spectral sum
BH26	$\lambda_k = k(k+25)$, $\Sigma_{10} = \sum_{k=1}^{10} \lambda_k = 1760$	KK eigenvalue ladder

The canonical convergence point is $\text{SSq} = 0.57$:

$$|\text{VDS}(0.57)| = \text{Li}_{26}(0.57) \approx 0.5700 \quad (< 1, \text{absolutely convergent})$$

2. VDS Variant Branch

2.1 Branch Definitions

The VDS branch introduces three derived quantities from the polylogarithm pair $(\mathrm{Li}_{25}, \mathrm{Li}_{26})$:

$$\mathrm{vds_prime} = \frac{d}{dz} \mathrm{Li}_{26}(z) \bigg|_{z=\mathrm{SSq}} = \frac{\mathrm{Li}_{25}(\mathrm{SSq})}{\mathrm{SSq}} \approx 1.0$$

$$\mathrm{vds_density} = \mathrm{Li}_{26}(\mathrm{SSq}) \times \rho_{\mathrm{SCm}} \quad \mathrm{J/m}^3$$

$$\mathrm{vds_k_weighted} = \mathrm{Li}_{25}(\mathrm{SSq}) + 25 \cdot \mathrm{Li}_{26}(\mathrm{SSq})$$

The BH26-weighted amplitude $\mathrm{vds_k_weighted}$ is the bridge quantity used in Stage 8 of the simulation engine (BSFG):

$$g_{\mathrm{BSFG}} = a_{\mathrm{DPM}} \times \mathrm{joint_coeff} \times \mathrm{vds_k_weighted} \times \frac{1}{\Sigma_{N=10} \{1760\}}$$

2.2 Physical Interpretation

The derivative $\mathrm{Li}_{25}/\mathrm{SSq} \approx 1$ demonstrates that $\mathrm{SSq} = 0.57$ is a **fixed-point** of the VDS calibration sensitivity: small perturbations in SSq produce proportionally equal changes in the series sum. This is the mathematical content of the "triple-convergence" condition.

3. DVP Variant Branch

3.1 Branch Definitions

The DVP (Dipole Vortex Prime) branch operates on primes $p > 26$ to encode the prime-vortex spectral sieve:

$$a(p) = \frac{\pi(p)}{p^{26}}, \quad \zeta_{\mathrm{DVP}}(\mathrm{SSq}) = \sum_{p > 26} a(p)$$

where $\pi(p)$ is the prime-counting function and p^{26} matches the VDS denominator exponent.

The dominant term at $p = 29$ (first prime above 26):

$$a(29) = \frac{0.57^{\pi(29)}}{29^{26}} = \frac{0.57^{10}}{29^{26}}$$

Branch quantities:

$$\zeta_{\mathrm{sum}} = \zeta_{\mathrm{DVP}}(0.57), \quad \mathrm{pair_product} = a(29) \times a(31), \quad \mathrm{spectral_floor} = a(p_{\mathrm{max}})$$

3.2 Physical Interpretation

The DVP prime sieve generates a **vorticity floor** for the DPM vacuum spiral. Each prime $p > 26$ above the 26-dimensional threshold contributes a quantised vortex mode. The p^{26} denominator enforces dimensional confinement to the 26-dimensional UQFF space.

4. BH26 Variant Branch (DH26)

4.1 Kaluza-Klein Spectral Ladder

The BH26 branch implements the Kaluza-Klein eigenvalue ladder on S^{25} :

$$\lambda_k = k(k + 25), \quad k = 1, 2, \dots, N$$

$$\Sigma_N = \sum_{k=1}^N \lambda_k = \frac{N(N+1)(2N+27)}{6} - N \cdot 25, \quad \Sigma_{10} = 1760$$

The Casimir vacuum energy from the ladder:

$$E_{\text{Casimir}} = \frac{\hbar f_{\text{RR}}}{2} \sum_{k=1}^N \frac{1}{\lambda_k} \quad [\text{J}]$$

where $f_{\text{RR}} = 1.15 \times 10^{14}$ Hz is the BH26 ring resonance frequency.

4.2 Topological Bridge to VDS

The BH26-to-VDS topological coupling:

$$\text{vds_coupling} = \sum_{k=1}^N \lambda_k^{-26}$$

This quantity bridges the spectral ladder to the VDS polylogarithm, completing the geometric closure of all three number systems at $\text{SSq} = 0.57$.

4.3 Degeneracy

The multiplicity on S^{25} at $k=1$: $\binom{26}{25} = 26$ (the 26-dimensional UQFF degeneracy factor).

5. VDS $\times$ DVP Coupled Field

5.1 Normalised Weights and Geometric-Mean Coupling

$$w_{\text{VDS}} = \frac{\text{Li}_{26}(\text{SSq})}{\text{Li}_{25}(\text{SSq})} \approx 1.0$$

$$w_{\text{DVP}} = \frac{\zeta_{\text{DVP}}(\text{SSq})}{a(29)}$$

$$\text{joint_coeff} = \sqrt{w_{\text{VDS}} \times w_{\text{DVP}}}$$

$$\text{variant_branch} = |w_{\text{VDS}} - w_{\text{DVP}}|$$

The geometric mean $\sqrt{w_{\text{VDS}} \times w_{\text{DVP}}}$ is the **field coupling** between the continuous-vacuum (VDS) and discrete-vortex (DVP) representations of the same underlying DPM vacuum geometry.

5.2 Triple-Point Condition

At $\mathrm{SSq} = 0.57$, simultaneous convergence (within 1%) holds:

$$\left| \frac{g_C - g_R}{g_R} \right| < 0.01, \quad \left| \frac{g_R - g_{\mathrm{BSFG}}}{g_{\mathrm{BSFG}}} \right| < 0.01, \quad \left| \frac{g_C - g_{\mathrm{BSFG}}}{g_{\mathrm{BSFG}}} \right| < 0.01$$

where g_C = Compressed-MUGE, g_R = Resonance-MUGE, g_{BSFG} = buoyancy gravity. This triple convergence is what determines $\mathrm{SSq} = 0.57$ uniquely (see PAPER_₁₁₅₄ for derivation).

6. BH26 $\times$ BSH Cross-Resonance

The BH26 $\times$ BSH resonance evaluates the Buoyancy Harmonic Series (PAPER_₄₂₉) at each spectral frequency bin of the BH26 ladder:

$$f_k = \frac{f_{\mathrm{RR}}}{\lambda_k} \quad [\mathrm{Hz}]$$

$$\mathrm{resonance}_k = \mathrm{BSH}(\omega_k) \cdot \cos(\pi t_n)$$

$$\mathrm{energy_density}_k = \mathrm{resonance}_k \times \rho_{\mathrm{SCm}} \quad [\mathrm{J/m}^3]$$

This cross-resonance provides the **frequency-domain bridge** between the discrete BH26 spectral ladder and the continuous SCm vacuum density field.

7. Test Suite: T61--T80

Test	Branch	Assertion
T61	VDS	$\mathrm{vds_prime} > 0$
T62	VDS	$\mathrm{vds_density} > 0$
T63	VDS	$\mathrm{vds_k_weighted} > 0$
T64	DVP	$\mathrm{zeta_sum} > 0$
T65	DVP	$n_{\mathrm{primes}} > 0$
T66	DVP	$a(29) > 0$
T67	BH26	$\Sigma_{10} = 1760$
T68	BH26	$\mathrm{casimir_energy} > 0$
T69	BH26	$\mathrm{degeneracy} = 26$
T70	BH26	$\mathrm{vds_coupling} > 0$
T71	VDS $\times$ DVP	$\mathrm{joint_coeff} > 0$
T72	VDS $\times$ DVP	$\mathrm{variant_branch} \geq 0$
T73	VDS $\times$ DVP	$w_{\mathrm{VDS}} > 0$
T74	VDS $\times$ DVP	$w_{\mathrm{DVP}} > 0$
T75	BSH Res.	$\mathrm{energy_density} \geq 0$
T76	BSH Res.	$\mathrm{freq_k} > 0$
T77	BSH Res.	$\mathrm{resonance} \geq 0$
T78	Coupled	$g_{\mathrm{BSFG}} = a_{\mathrm{DPM}} \times \mathrm{joint_coeff} \times \mathrm{vds_k_weighted}$
T79	Convergence	$ \mathrm{vds_prime} - 1 < 0.1$
T80	Integration	All branch results non-zero simultaneously

Result: 80/80 tests PASS (commits 6e69e2f5, dad3ae0d)

8. CP4 Calculator Classes

Session 202 produced five CP4 Phase-4 calculator classes (CP4 \#644--\#648):

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| \# | Class | PAPER |
|----|-----|-----|
| 644 | VDSBranchCalculator | PAPER\_1151 |
| 645 | DVPBranchCalculator | PAPER\_1151 |
| 646 | BH26BranchCalculator | PAPER\_1151 |
| 647 | VDSDVPCoupledCalculator | PAPER\_1151 |
| 648 | BH26BSHResonanceCalculator | PAPER\_1151 |
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9. Conclusions

Phase H202 completes the branch-level derivation of all three UQFF number systems (VDS, DVP, BH26) and their coupled-field geometry. The geometric-mean joint coupling coefficient $\sqrt{w_{\text{VDS}} \times w_{\text{DVP}}}$ is the fundamental UQFF calibration quantity for the BSFG gravity path. All results are verified at $\mathrm{SSq} = 0.57$.

References

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Updated: 2026-05-06 (Session 234, PAPER_1151). Compliant: CVW v2.0.0, G6 SM Anchor Gate.
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